PACKGE INSERT

GRANISETRON KABI 1 MG/ML

Concentrate Solution for Injection/Infusion

CONTAINS:
Each ml of Granisetron Kabi concentrate solution for injection/infusion contains 1 mg of granisetron (as the hydrochloride). Other ingredients are citric acid (monohydrate), hydrochloric acid, sodium chloride, sodium hydroxide, water for injections.

DESCRIPTION:
Granisetron Kabi 1 mg/ml is a sterile, colourless solution, containing 1 mg/ml Granisetron in 1 ml and 3 ml isotonic aqueous solution. The diluted solution for injection and infusion should be clear and free from particles. Granisetron Kabi 1 mg/ml can also be diluted in 20 to 50 ml compatible infusion fluid and then given over five minutes as an intravenous infusion in any of the following solutions:

- 0.9 % w/v sodium chloride injection
- 5 % w/v glucose injection
- Lactated Ringer's Solution;

No other diluents should be used.

PHARMACOLOGICAL PROPERTIES

PHARMACODYNAMIC PROPERTIES:
Pharmacotherapeutic group (ATC code): Serotonin (5-HT3) antagonists (A04AA02).
Granisetron is a potent anti-emetic and highly selective antagonist of 5-hydroxytryptamine (5-HT3) receptors. Radioligand binding studies have demonstrated that granisetron has negligible affinity for other receptor types, including 5-HT and dopamine D2 binding sites. Granisetron is effective intravenously, either prophylactically or by intervention, in abolishing the retching and vomiting evoked by administration of cytotoxic drugs or by whole body X-irradiation.

PHARMACOKINETICS:
General characteristics

Absortion
Following intravenous doses in the range of 20-160 µg/kg, plasma pharmacokinetics (Cmax and AUC) were generally dose-proportional in both healthy subjects and in patients receiving chemotherapy. The mean plasma half-life was 5.2 h in healthy subjects and 8.7 h in patients receiving chemotherapy.

Distribution
Granisetron is extensively distributed, with a mean volume of distribution of approximately 3l/kg; plasma protein binding is approximately 65%.

Biotransformation
Biotransformation pathways involve N-demethylation and aromatic ring oxidation followed by conjugation.

Elimination
Clearance is predominantly by hepatic metabolism. Urinary excretion of unchanged granisetron averages 12% of dose whilst that of metabolites amounts to about 47% of dose. The remainder is excreted in faeces as metabolites. Mean plasma half-life in patients is approximately nine hours, with a wide inter-subject variability.

Characteristics in patients
The plasma concentration of granisetron is not clearly correlated with the anti-emetic efficacy. Clinical benefit may be conferred even when granisetron is not detectable in plasma.

In elderly subjects after single intravenous doses, pharmacokinetic parameters were within the range found for non-elderly subjects.

In children, after a single IV dose, the pharmacokinetics resembles that of adults when relevant parameters (volume of distribution, plasma clearance) are adjusted for body weight.

In patients with severe renal failure, data indicate that pharmacokinetic parameters after a single intravenous dose are generally similar to those in normal subjects.

In patients with hepatic impairment due to neoplastic liver involvement, total plasma clearance of an intravenous dose was approximately halved compared to patients without hepatic involvement. Despite these changes, no dosage adjustment is necessary**.**

INDICATION:
Indicated for the prevention and treatment (control) of
a) Acute and delayed nausea and vomiting associated with chemotherapy and radiotherapy.
b) Post-operative nausea and vomiting.

POSOLOGY AND METHOD OF ADMINISTRATION:
Posology

Chemo- and radiotherapy-induced nausea and vomiting (CINV and RINV)
Prevention (acute and delayed nausea)

A dose of 1-3 mg (10-40 µg/kg) of Granisetron Kabi should be administered either as a slow intravenous injection or as a diluted intravenous infusion 5 minutes prior to the start of chemotherapy. The solution should be diluted to 5 ml per mg.

Treatment (acute nausea)

A dose of 1-3 mg (10-40 µg/kg) of Granisetron Kabi should be administered either as a slow intravenous injection or as a diluted intravenous infusion and administered over 5 minutes. The solution should be diluted to 5 ml per mg. Further maintenance doses of Granisetron Kabi may be administered at least 10 minutes apart. The maximum dose to be administered over 24 hours should not exceed 9 mg.

Combination with adrenocortical steroid

The efficacy of parenteral granisetron may be enhanced by an additional intravenous dose of an adrenocortical steroid e.g. by 8-20 mg dexamethasone administered before the start of the cytostatic therapy or by 250 mg methyl-prednisolone administered prior to the start and shortly after the end of the chemotherapy.

Paediatric population

The safety and efficacy of Granisetron Kabi in children aged 2 years and above has been well established for the prevention and treatment (control) of acute nausea and vomiting associated with chemotherapy and the prevention of delayed nausea and vomiting associated with chemotherapy. A dose of 10 – 40 µg/kg body weight (up to 3 mg) should be administered as an i.v. infusion, diluted in 10 to 30 ml infusion fluid and administered over 5 minutes prior to the start of chemotherapy. One additional dose may be administered within a 24 hour-period if required. This additional dose should not be administered until at least 10 minutes after the initial infusion.

Post-operative nausea and vomiting (PONV)

A dose of 1 mg (10µg/kg) of Granisetron Kabi should be administered by slow intravenous injection. The maximum dose of Granisetron Kabi to be administered over 24 hours should not exceed 3 mg.

For the prevention of PONV, administration should be completed prior to induction of anaesthesia.

Paediatric population

Currently available data are described in section 5.1. but no recommendation on a posology can be made. There is nsufficient clinical evidence to recommend administration of the solution for injection to children in prevention and treatment of Post-operative nausea and vomiting (PONV).

Special populations
Elderly and renal impairment

There are no special precautions required for its use in either elderly patients or those patients with renal or hepatic impairment.

Hepatic impairment

There is no evidence to date for an increased incidence of adverse events in patients with hepatic disorders. On the basis of its kinetics, whilst no dosage adjustment is necessary, granisetron should be used with a certain amount of caution in this patient group (see section 5.2.Pharmacokinetic properties).

Method of administration

Administration may be as either a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 ml infusion fluid and administered over 5 minutes.

ROUTE OF ADMINISTRATION: INTRAVENOUS

CONTRAINDICATIONS:

- If you are allergic (hypersensitive) to granisetron or any of the other ingredients of this medicine.
- In children under 2 years of age because insufficient experience is available.

Do not have Granisetron Kabi if the above applies to you. If you are not sure talk to your doctor, nurse or pharmacist before having Granisetron Kabi.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE:
Granisetron may reduce intestinal motility. Patients showing symptoms of sub-acute intestinal obstruction following administration of Granisetron Kabi should be monitored carefully.

5-HT3 antagonists, such as granisetron, may be associated with arrhythmias or ECG abnormalities. This potentially may have clinical significance in patients with pre-existing arrhythmias or cardiac conduction disorders or patients who are being treated with antiarrhythmic agents or beta-blockers.

No special precautions are required for the elderly or renally and/or hepatically impaired patient. Although to date no signs of an increased incidence of adverse events have been observed in hepatically impaired patients, owing to the kinetics a degree of caution should be exercised in using granisetron with this category.

This medicinal product contains 1.37 mmol sodium (or 31.5 mg) per maximum daily dose of 9 mg. To be taken into consideration by patients on a controlled sodium diet.

INTERACTIONS WITH OTHER MEDICINAL PRODUCTS AND OTHRE FORMS OF INTERACTION:
No definitive drug-drug interaction study has been performed. Granisetron is primarily metabolised by CYP3A enzymes and does not induce or inhibit any other CYP enzymes.

In vitro, it could be shown that metabolism of granisetron is inhibited by ketoconazole, a potent CYP3A inhibitor. Coadministration of granisetron with systemic ketoconazole may, therefore, increase granisetron's elimination half-life.

In humans, hepatic enzyme induction with phenobarbital resulted in an increase in total plasma clearance of granisetron of approximately 25 %. The clinical significance of this change is not known.

So far no signs of interaction have been observed between granisetron and medicinal products that are often prescribed in anti-emetic therapy, such as benzodiazepines, neuroleptics and anti-ulcer medications.

Granisetron injections have shown no apparent drug interaction with emetogenic cancer chemotherapies.

No specific interaction studies have been conducted in anaesthetised patients, but granisetron injections have been safely administered with commonly used anaesthetic and analgesic agents.

PREGNANCY AND LACTATION:
Pregnancy

Whilst animal studies have shown no teratogenic effects, there is no experience with the use of granisetron during human pregnancy. Therefore Granisetron Kabi should not be administered to women who are pregnant unless there are compelling clinical reasons.

Lactation

There are no data on the excretion of granisetron in breast milk. Breast feeding should therefore be discontinued during therapy with Granisetron Kabi

UNDESIRABLE EFFECTS:
The most frequent adverse effect is headache, occurring in about 14% of patients. Other less common adverse events associated with granisetron administration include hypersensitivity reactions (e.g. anaphylaxis), constipation, diarrhoea, asthenia and somnolence.

Incidence and severity are given in the following table:

Cardiac disorders	*Rare:* arrhythmias such as sinus bradycardia, atrial fibrillation, varying degrees of AV-block, ventricular ectopy (including non-sustained tachycardia), ECG abnormalities
Nervous system disorders	*Very common:* headache *Common:* somnolence, agitation, anxiety, insomnia, taste disorder.

	Rare: dystonia and dyskinesia have been reported with medicines in the 5-HT3 antagonist class.
Eye disorders	*Uncommon:* abnormal vision
Ear and labyrinth disorders	*Common:* dizziness
Gastrointestinal disorders	*Common:* diarrhoea, constipation, anorexia
Skin and subcutaneous tissue disorders	*Uncommon:* skin rashes *Rare:* local irritations at administration site after repeated intravenous administration
Vascular disorders	*Common:* hypertension *Rare:* hypotension
General disorders	*Common:* fever, asthenia
Immune system disorders	*Rare:* hypersensitivity reactions, sometimes severe (e.g., anaphylaxis, shortness of breath, hypotension, urticaria) *Very rare:* oedema (including facial oedema)
Hepatobiliary disorders	*Rare:* abnormal hepatic function, raised transaminase levels

SYMPTOMS AND TREATMENT OF OVERDOSE:
Overdosage of up to 30 mg of granisetron injection (10 times the recommended dose) has been reported without symptoms or only the occurrence of a slight headache. There is no specific antidote for granisetron overdosage. In case of overdosage, symptomatic treatment should be given.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE:
Somnolence is a common side effect observed after granisetron treatment. Depending on the patient's individual reaction this may impair his/her ability to drive, to operate machinery or to work at high altitude. If the patient feels drowsy after treatment with Granisetron Kabi he/she should be advised not to drive, not to operate machinery and not to carry out any work that requires safe foothold.

STORAGE CONDITION:
Store below 30°C. Keep the ampoules in the outer carton in order to protect from light. Do not freeze.

STORAGE AFTER DILUTION:
- Once diluted Granisetron Kabi should be used immediately
- Chemical and physical in-use stability has been demonstrated for 24 hours at 25°C.
- If not used immediately, the ready to use solution should be stored at 25°C, protected from sunlight and used within 24 hours.

SHELF LIFE OF THE FINISHED MEDICINAL PRODUCT:
3 years.
Once opened the product should be used immediately.

Ideally, intravenous infusions of Granisetron Kabi should be prepared at the time of administration. After dilution, or when the container is opened for the first time, the shelf life is 24 hours when stored at ambient temperature (25°C) in normal indoor illumination protected from direct sunlight. It must not be used after 24 hours. If to be stored after preparation, Granisetron Kabi infusions must be prepared under appropriate aseptic conditions.

Granisetron 1 mg / ml is compatible with Dexamethason dihydrogenphosphate dinatrium in a concentration of 10-60 µg/ml of Granisetron and 80-480 µg/ml Dexamethasonphosphate diluted in sodium chloride 0.9 % or Glucose5 % solution over a period of 24 hours.

Any unused product or waste material should be disposed of in accordance with local requirements.

PACK SIZES:
The pack may contain 5 or 10 clear glass ampoules. The ampoules contain either 1 ml or 3 ml Granisetron Kabi 1 mg/ ml concentrate solution for injection/infusion. Not all pack sizes may be marketed.

OTHER INFORMATION:
The following information is intended for medical healthcare professionals only:

Instructions for dilution:

Dilute before use. For single use only. Any unused portion should be discarded.

The diluted injections and infusions are to be inspected visually for particulate matter prior to administration. They should only be used if the solutionis clear and free from particles.

Adults: The contents of a 1 ml ampoule can be diluted to a volume of 5 ml; the contents of a 3 ml ampoule can be diluted to a volume of 15 ml.

Granisetron Kabi can also be diluted in 20 to 50 ml compatible infusion fluid and then given over five minutes as an intravenous infusion in any of the following solutions:

0.9 % w/v sodium chloride injection
5 % w/v glucose injection
Lactated Ringer's Solution

No other diluents should be used.

Children 2 years of age and older: To prepa re the dose of 20 - 40 µg/kg, the appropriate volume is withdrawn and diluted with infusion fluid (as for adults) to a total volume of 10 to 30 ml.

As a general precaution, Granisetron Kabi should not be mixed in solution with other drugs.

MANUFACTURER:
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PRODUCT REGISTRATION HOLDER:
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